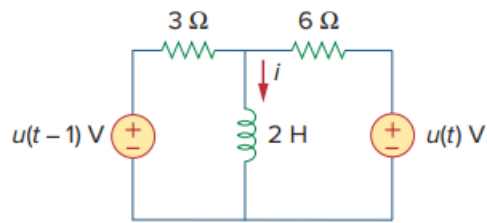


Problem

For the circuit in Fig. 7.127, calculate $i(t)$ if $i(0) = 0$.

Fig. 7.127:



Step-by-step solution

Step 1 of 6

Refer to Figure 7.127 in the textbook.

Consider the following relations for the input signals:

$$u(t) = \begin{cases} 1, & t \geq 0, \\ 0, & t < 0. \end{cases}$$

$$u(t-1) = \begin{cases} 1, & t \geq 1, \\ 0, & t < 1. \end{cases}$$

Draw the circuit for $t < 0$.

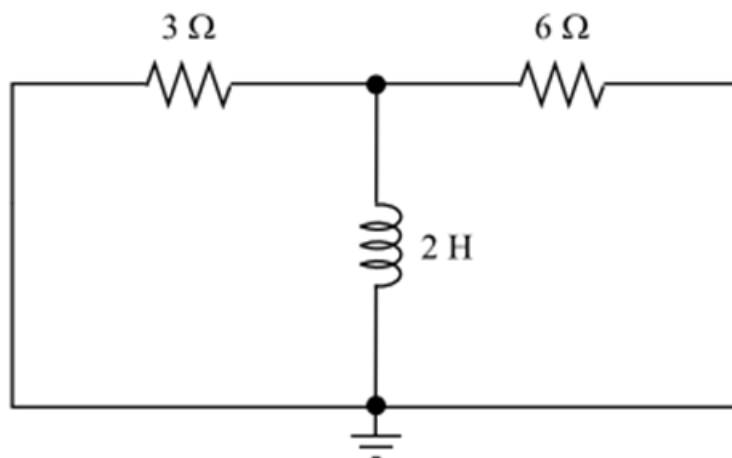


Figure 1

Step 2 of 6

Write the general expression for the current through the inductor in the circuit.

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-\frac{t}{\tau}}$$

Here, $i(0)$ is the initial value of inductor current, $i(\infty)$ is the steady state value of inductor current and τ is the time constant.

Consider that the initial value of the current is zero. Therefore, $i(0) = 0$.

Find the time constant of the circuit by replacing the voltage sources with their equivalent impedances. The equivalent resistance in the circuit looking into the inductor is,

$$\begin{aligned} R_{eq} &= 3\ \Omega \parallel 6\ \Omega \\ &= \frac{(3\ \Omega)(6\ \Omega)}{3\ \Omega + 6\ \Omega} \\ &= \frac{18}{9} \\ &= 2\ \Omega \end{aligned}$$

The time constant of RC circuit is,

$$\tau = \frac{L}{R_{eq}}$$

Substitute $2\ \Omega$ for R_{eq} and $2\ \text{H}$ for L .

$$\begin{aligned} \tau &= \frac{2\ \text{H}}{2\ \Omega} \\ &= 1\ \text{s} \end{aligned}$$

Step 3 of 6

For $0 < t < 1$, the source, $u(t)$ exists and $u(t-1) = 0$. The circuit is in steady state and inductor acts as short circuit in steady state. Therefore, the $3\ \Omega$ resistance is eliminated due to short across it and the only resistance available in the circuit is $6\ \Omega$.

Draw the circuit for $0 < t < 1$.

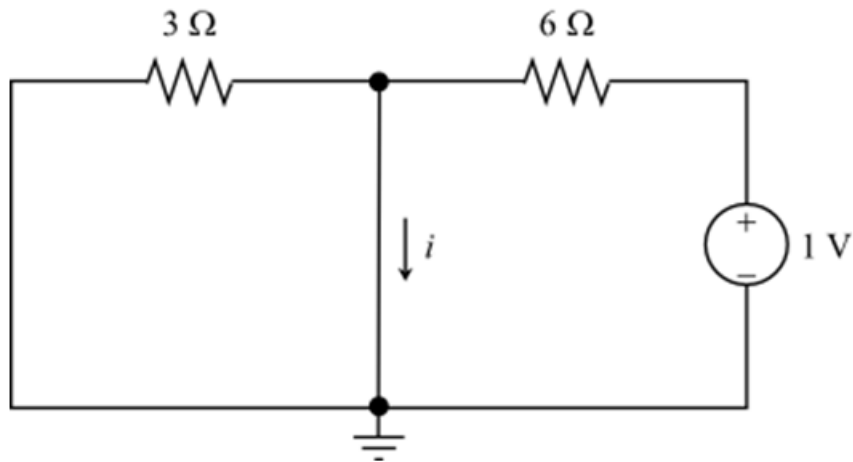


Figure 2

Step 4 of 6

Find the value of final current in the circuit.

$$\begin{aligned} i(\infty) &= \frac{u(t)}{R} \\ &= \frac{1}{6} \text{ A} \end{aligned}$$

Find the expression for the current in the circuit for $0 < t < 1$.

$$\begin{aligned} i(t) &= i(\infty) + [i(0) - i(\infty)]e^{-\frac{t}{\tau}} \\ &= \frac{1}{6} + \left[0 - \frac{1}{6}\right]e^{-\frac{t}{1}} \\ &= \frac{1}{6}(1 - e^{-t}) \text{ A} \end{aligned}$$

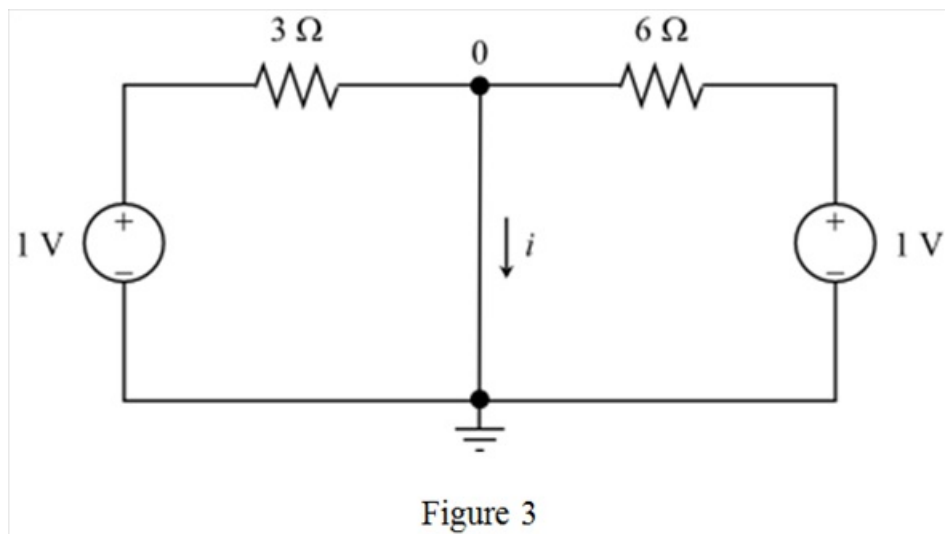
Find the value of current at $t = 1 \text{ s}$.

$$\begin{aligned} i(1) &= \frac{1}{6}(1 - e^{-1}) \\ &= \frac{1}{6}(1 - 0.368) \\ &= \frac{0.632}{6} \\ &= 0.105 \text{ A} \end{aligned}$$

Step 5 of 6

For $t > 1$, both the sources exist. The circuit is in steady state and inductor acts as short circuit in steady state.

Draw the circuit for $t > 1$.



Step 6 of 6

Write the general expression for the current through the inductor in the circuit.

$$i(t) = i(\infty) + [i(1) - i(\infty)] e^{-\frac{t-1}{\tau}}$$

Find the value of final current in the circuit.

$$\begin{aligned} i(\infty) &= \frac{0-1}{3} + \frac{0-1}{6} \\ &= -\frac{1}{3} - \frac{1}{6} \\ &= -\frac{1}{2} \\ &= -0.5 \end{aligned}$$

Find the expression for the current in the circuit.

$$\begin{aligned} i(t) &= i(\infty) + [i(1) - i(\infty)] e^{-\frac{t-1}{\tau}} \\ &= -0.5 + [0.105 - (-0.5)] e^{-\frac{t-1}{1}} \\ &= -0.5 - 0.395 e^{-(t-1)} \text{ A} \end{aligned}$$

Thus, the final expression for the current in the circuit is,

$$i(t) = \begin{cases} \frac{1}{6}(1 - e^{-t}) \text{ A}, & 0 < t < 1, \\ -0.5 - 0.395 e^{-(t-1)}, & t > 1. \end{cases}$$